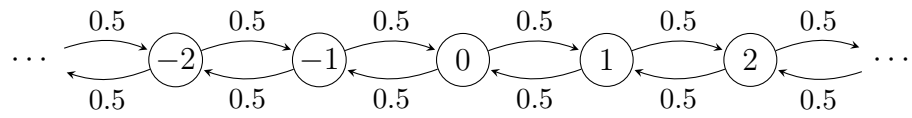


## IE 203 PS 10 – Solutions

### Q1.

$$S = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

**State Diagram:**



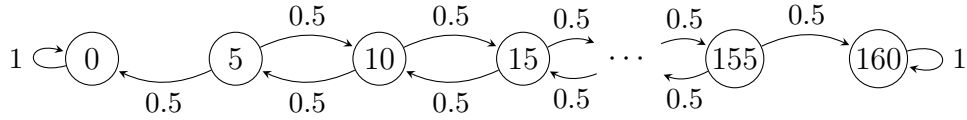
**One step transition probabilities:**

$$P(X_{t+1} = j \mid X_t = i) = \begin{cases} \frac{1}{2} & \text{if } j = i + 1 \\ \frac{1}{2} & \text{if } j = i - 1 \\ 0 & \text{for other } j \end{cases}$$

## Q2.

$X_t$ : The number of Kopecks that Alexei has.  $X_t \in S$ ,  $S = \{0, 5, 10, \dots, 160\}$ .

**State Diagram:**



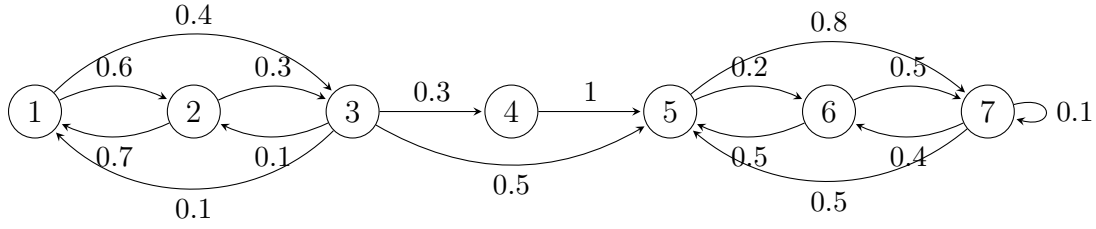
**One step transition probabilities:**

$$P(X_{t+1} = j \mid X_t = i) = \begin{cases} 0.5 & \text{if } i \in S \setminus \{0, 160\} \text{ and } j = i \pm 5 \\ 1 & \text{if } i = j = 0 \\ 1 & \text{if } i = j = 160 \\ 0 & \text{for other } i, j \end{cases}$$

**Initial State:**  $X_0 = 60$ .

### Q3.

a.



b.

- 1 is reachable from 2, 2 is reachable from 1, so 1 and 2 are communicating.
- $P_{13}^{(1)} > 0$  and  $P_{31}^{(1)} > 0$ , thus 1 and 3 are communicating.
- 1 and 2 are communicating and 1 and 3 are communicating, thus 2 and 3 are communicating.

Similarly, 5, 6 and 7 communicate with each other. State 4 does not communicate with any other state.

$$\text{Class 1} = \{1, 2, 3\}$$

$$\text{Class 2} = \{4\}$$

$$\text{Class 3} = \{5, 6, 7\}$$

c. Starting from state 1, the process eventually hits 4 or 5 and never revisits 1. Thus, 1 is transient. The states in the same class have to be of the same type, therefore 2 and 3 are also transient.

Starting from 4, the process never revisits 4. 4 is transient.

Starting from 5, the process will revisit 5 infinitely many times. 5, 6, 7 are recurrent.

d.  $P_{11}^{(2)} = ?$

$$\begin{aligned}
 P_{11}^{(2)} &= p_{12}p_{21} + p_{13}p_{31} + p_{14}p_{41} + p_{15}p_{51} + p_{16}p_{61} + p_{17}p_{71} + p_{11}p_{11} \\
 &= 0.6 \times 0.7 + 0.4 \times 0.1 \\
 &= 0.46
 \end{aligned}$$

e.  $P_{14}^{(3)} = ?$

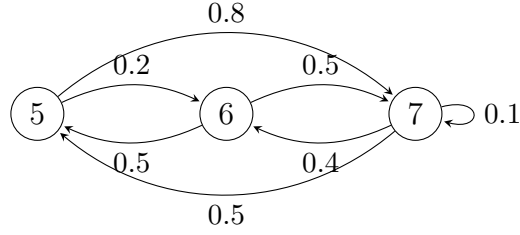
$$P_{14}^{(3)} = P_{13}^{(2)} \cdot p_{34}$$

$$P_{13}^{(2)} = p_{12}p_{23} \quad (\text{one possible path})$$

$$\begin{aligned}
 P_{14}^{(3)} &= p_{12} p_{23} p_{34} \\
 &= 0.6 \times 0.3 \times 0.3 = 0.054
 \end{aligned}$$

f. Regardless of the initial state, the process is absorbed by Class 3. This class is itself an ergodic and irreducible discrete-time Markov chain (will see at part g). Therefore, unique steady state probabilities exist.

g. Markov chain induced by Class 3:



$$\text{Period of 5} = \gcd\{n \in \mathbb{N}_+ : p_{55}^{(n)} > 0\} = \gcd\{2, 3, 4, \dots\} = 1$$

Thus, this MC is aperiodic. All states are recurrent as we have shown. Thus, this MC is ergodic.

We have shown all states  $\{5, 6, 7\}$  communicate with each other. Therefore, this MC is irreducible.

This MC is irreducible and ergodic, therefore unique steady state probabilities exist.  
**h.**

$$[\pi_5 \ \pi_6 \ \pi_7] = [\pi_5 \ \pi_6 \ \pi_7] \begin{bmatrix} 0 & 0.2 & 0.8 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.4 & 0.1 \end{bmatrix}$$

$$\pi_5 = 0.5 \pi_6 + 0.5 \pi_7$$

$$\pi_6 = 0.2 \pi_5 + 0.4 \pi_7$$

$$\pi_7 = 0.8 \pi_5 + 0.5 \pi_6 + 0.1 \pi_7$$

One of them is redundant. Using  $\pi_5 + \pi_6 + \pi_7 = 1$ :

$$\boldsymbol{\pi} = \begin{bmatrix} \underbrace{\frac{14}{42}}_{\pi_5} & \underbrace{\frac{10}{42}}_{\pi_6} & \underbrace{\frac{6}{14}}_{\pi_7} \end{bmatrix}$$

Due to the structure of the MC:  $\pi_1 = \pi_2 = \pi_3 = \pi_4 = 0$ .

★ We were able to “reduce” the whole MC due to its structure. For example, steady state probabilities don’t exist in the MC to the right, since initial state matters.

